# PHYSICS

**Paper 5054/32**
**Practical Test 32**

## General comments

**Question 4** was particularly challenging. Discharging of a capacitor through a resistor had not been set before, so the situation was new to the candidates. Also the discharge led to a curved graph of potential difference against time and candidates had to find the gradient of this graph and take data from the graph. However the question worked very successfully and produced good discrimination between candidates. More able candidates coped well with the question and less able candidates were able to use their graph plotting skills to obtain some marks.

## Comments on specific questions

***Section A***

**Question 1**

**(a)** Most candidates gained credit for the value of *L*. Virtually all candidates obtained a value in the correct range. The common mistakes made were the omission of the unit or not quoting the value to the nearest mm.

**(b)** Good answers used the 2 set squares like the jaws of vernier callipers with one edge of the set square against the edge of the rule. Diagrams that did not score the mark included;

- those where the set square was standing vertically on the rule so that it was not clear that the face of the set square was perpendicular to the line of spheres,
- those where the hypotenuse of the set square was used at the end of the line of spheres because it was not clear that the edge of the set square was perpendicular to the line of spheres.

**(c) (i)** Most candidates correctly determined the value *d* from their value of *L*.

**(ii)** The question asked the candidates to determine the average mass *m* of a sphere. It was expected that the candidates would measure the total mass of all 20 spheres that they had available and then divide by 20 to obtain the average mass of a sphere. In the event quite a number of candidates just measured the mass of 1 sphere. Two other common mistakes in this section were to:

- give the mass *m* as the total mass of the 20 spheres,
- not subtract the mass of the container from the total mass that had been measured on the top-pan balance.

**(iii)** At most Centres the spheres used must have been very uniform, because even those candidates who only measured the mass of one sphere obtained a good value for the density of the material of the sphere.


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**Question 2**

**(a)** It was good to see that the majority of candidates had followed the instructions and had the start position of the mass hanger just below the pulley. Most candidates had a value for *y* that was in excess of 60 cm, with only a few candidates having low values such as 20 cm or 30 cm. The normal requirement for measurements to the nearest mm was relaxed here, so that Examiners could credit those candidates who used a large height through which the mass would fall but omitted to quote the measurement to the nearest mm.

**(b)** The best diagrams showed the set square between the vertical metre rule and the horizontal floor. Other diagrams only showed the metre rule and the set square without the floor; such diagrams did not score the mark.

**(c) (i)** The question asked the candidates to record the total mass of the mass hanger and the slotted masses. However it was clear that some candidates had also included the mass of the wooden block or the additional mass that was on top of it. Using the apparatus specified in the instructions, it should have been impossible to obtain a mass greater than 100 g.

**(ii)** The question asked candidates to determine the average time for the mass hanger to fall to the floor. Candidates were therefore expected to take at least 2 measurements and find the average. Quite a number of candidates took only 1 measurement. It was pleasing to see that fewer candidates than usual had systematic errors (e.g. 0.0250 s) or measured the time to the nearest second.

**(d)** The most common errors in the calculation of the acceleration were;

- carrying out the calculation in cm and putting the unit as $m/s^2$,
- using a unit of m/s or cm/s.

**Question 3**

**(a)** As in the previous two questions, the use of the word 'average' in a question should imply that more than one measurement is to be made. However in this question, not many candidates realised that more than one measurement of the diameter of the cross-wire object was required.

**(b) (i)** It was expected that Centres would use a 15 cm focal length lens in this experiment. With an object distance of 20 cm, such a lens should give an image distance of 60 cm, which was the expected value for *v*. A number of candidates reported that they were unable to obtain an image and had increased the object distance (*u*) without consulting the Supervisor. In other cases the Supervisor reported that there were insufficient 15 cm focal length lenses and that 20 cm focal length lenses had been used. In either event the candidate was given credit for a reasonable answer. The allowed range for *v* was from 45 cm to 80 cm, which allowed both 15 cm and 20 cm focal length lenses to be used, correspondingly the allowed range for *D* was from 4.0 cm to 9.0 cm.

**(ii)** Despite the large range of values allowed for *D*, a large number of candidates did not gain credit because only 1 measurement was taken.

**(c)** Virtually all candidates gained credit for the correct calculation of *m* and *f*, partly because the units given for *m* were ignored. *m* should, of course, have had no units, but many candidates gave the unit as cm. Where it was clear that a 15 cm focal length lens had been used the range for the focal length was from 13.0 cm to 17.0 cm as in the published mark scheme. However if it was clear that a 20 cm focal length lens had been used, then a range of 18.0 cm to 22.0 cm was used.


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***Section B***

**Question 4**

**(a)** Many candidates were given credit for the circuit diagram without any problem. Essentially the resistor, capacitor and power supply should have been connected in parallel. The omission of a switch was ignored because it was felt that at some Centres disconnecting two of the connecting leads would have been used as a switch. However if a switch was included in the circuit it was important that it was between the power supply and the capacitor rather than the resistor and the capacitor. In the latter case no credit was given for the circuit because it would not have worked i.e. the capacitor would have remained connected to the power supply and would not have discharged. Apart from switch positioning errors, the only other error that candidates made on the circuit diagram was connecting the power supply, resistor and capacitor in series rather than in parallel.

**(b)** Most candidates obtained a correct value for $V_0$, the most common errors were:

- not quoting the potential difference to 0.1 V or better, e.g. 3 V,
- omitting the unit from the quantity.

**(c)** Most candidates obtained a correct value for $V_R$ which was less than or equal to the value of $V_0$. The errors were as stated in **(b)** but were not penalised if they had already been penalised in **(b)**.

**(d)** Some Supervisors complained that candidates had insufficient time for this experiment. This could only be the case if readings were not done continuously, i.e. the capacitor was discharged for 10 seconds and the final voltage was recorded, the capacitor was recharged and then discharged for 20 seconds, etc. It was anticipated that candidates would take readings continuously as they would do if they were, for example, plotting a cooling curve. In this experiment the switch should have been opened and the stopwatch started, readings on the voltmeter should then have been taken at 10 s, 20 s, etc. Candidates did not receive credit if they:

- omitted units from the headings of the table,
- did not start the stopwatch and open the switch at the same time; this led to an apparent large drop in voltage in the first few seconds. This was penalised as a systematic error,
- only did readings every 20 seconds resulting in only 5 points for the graph.

**(e)** Graph plotting skills seemed to have improved with many candidates getting full marks for the graph. Examiners were slightly more lenient with the suitable scale mark than in previous years. First of all the graph was allowed to start at the origin. With the total time being 80 seconds a horizontal scale of 1 cm = 5 s, was most appropriate, but 1 cm = 10 s was allowed as this just filled half the page. Equally, with a maximum voltage in the region of 3.0 V, a vertical scale of 1 cm = 0.2 V would have been appropriate, but a vertical scale of 1 cm = 0.25 V was allowed as again this just filled half the page.

**(f)** Only the most able candidates gained credit in **(f)** and **(g)**. Firstly many candidates interpreted $V_R = 0.5\ V_0$ as $V_R = 0.5$ V, and hence measured the gradient at 0.5 V. Some candidates did not draw a tangent but used two points on the curve, either side of the point at which the gradient was required. This is not an acceptable method. Others drew a tangent in the correct place but only used a very small triangle to determine the gradient, this lost the large triangle mark. Finally the gradient was often quoted to 2 decimal places rather than 2 significant figures, e.g. 0.04 (V/s), hence the correct calculation to 2/3 significant figures mark was lost.

**(g)** As in **(f)** $V_R = 0.37\ V_0$ was often interpreted as $V_R = 0.37$ V, so candidates obtained much higher values for the time than expected.


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